

From the labs

The folks at IIT Kharagpur are working wonders with the intelligence at their disposal.

Space age

Space research can yield some of the most fascinating results with even more fascinating applications.

Greatest unsolved problems in science

Mankind has achieved a lot, but some mysteries remain beyond our reach.

Ronak Gupta

readersletters@didgit.in

In the last two centuries, Science has answered many questions about nature and the laws that govern it. We've been able to research galaxies and the atoms that make up matter. We've built machines that can compute and solve problems that no human can work out. We've solved age-old problems in Mathematics and created theories that gave maths new problems. This article, however, is not about these achievements. It's about those problems in science that have left scientists scratching their heads, wondering if they'll ever yell, "Eureka!"

Turbulence

Turbulence isn't a new word. You're probably more familiar with it as the word that describes the sudden bumpiness on an aircraft's. However, turbulence in fluid mechanics is a whole different ball game. Flight turbulence, more technically referred to as 'clear-air turbulence', is caused due to the meeting of two bodies of air moving at different speeds. Physicists, however, have a much tougher time explaining this phenomena of turbulence in fluids. Mathematicians have nightmares about it.

Fluid turbulence is all around us. The jet from a tap that's turned on completely breaks into chaotic patches of fluid, different from the unified stream of jet we get when we only half open it. This is one of the classical examples of turbulence and is used to explain it to school students and graduates alike. Turbulence is ubiquitous



A meteor shower could have seeded life on earth, according to the panspermia theory

in nature, occurring in various geophysical and oceanic flows. It's also of importance to engineers as it occurs frequently in flow over turbine blades, aerofoils and other bodies. Turbulence is characterised by random fluctuations in variables like velocity and pressure.

While there have been plenty of experiments and empirical data regarding turbulence, we're still far from a convincing theory about what precisely triggers turbulence in a fluid, how it is controlled, and what exactly lends it order through chaos. Making the problem even more difficult to tackle is the fact that the equations governing the motion of a fluid – namely the Navier-Stokes equations – are notoriously hard to analyse. Scientists resort to high performance computing

techniques, along with experiments and theoretical simplifications to study the phenomena, but a complete theory of turbulence is still lacking, making fluid turbulence one of the most important unsolved problems in physics today. In fact, Nobel laureate, Richard Feynman, declared it as "the most important unsolved problem in classical physics." Quantum physicist Werner Heisenberg, when asked what he would ask God if given the opportunity, remarked: "I'll ask him two questions. Why relativity? And why turbulence? I really believe he will have an answer to the first."

We got a chance to speak to Padma Vibushan recipient, Prof. Roddam Narasimha and this is what he had to say: "To this day we are unable to predict the